INCH-POUND

MIL-DTL-27893F (GL)
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SUPERSEDING
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10 April 1989

DETAIL SPECIFICATION

HARNESSES, PARACHUTIST'S INDIVIDUAL EQUIPMENT PACK ("H"- TYPE)

This specification is approved for use by the Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of general purpose harness, having an "H" design and made of nylon webbing. The harness is equipped with quick-fit adapters, buckles and snaps to secure it around the load and to a parachute harness worn by the parachutist.

1.2 Classification. The harness shall be of the following type:

Type I - Deleted (see 6.3)
Type II - Harness, Single Point Release

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Attn: SSCNC-UTE Natick, MA 01760-5017 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1670

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been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Drawings. The following Government drawings forms a part of this document to the extent specified herein. Unless otherwise specified, the issue is that cited in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

11-1-3008 - Harness, Parachutist's Individual Equipment Pack; Single Point Release, Type II, Assembly
11-1-3009 - Harness
11-1-3010 - Strap, Leg, Release Assembly
11-1-3011 - Handle Release Assembly
11-1-3012 - Strap, Harness, Attaching

(Copies of drawings are available from the U.S. Army Natick Research, Development and Engineering Center, ATTN: AMSSC-S-IMI, Natick, MA 01760-5017).

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Conformance inspection. The end item, materials and components shall be subjected to conformance inspection in accordance with 4.3.

3.3 Materials and components. Materials and components shall conform to the requirements on the applicable drawings, to subsidiary specifications and standards applicable thereto, and as specified herein.

3.3.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or

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environmentally preferable materials should be used to the maximum extent possible provided the material meets or exceeds the operational requirements, and promotes economically advantageous life cycle costs.

3.4 Construction. The construction shall conform in all respects to the drawings listed in section 2 and all subsidiary drawings pertaining thereto, and as specified herein.

3.4.1 Stitching, machine. All machine stitching shall be as specified on the applicable drawings and as specified herein.

3.4.1.1 Types 301 and 308 stitching. Ends of all type 301 stitching shall be backstitched or overstitched not less than 1/2 inch in accordance with the required stitch pattern. Ends of type 308 stitching shall be secured by crowding a minimum of three stitches (accomplished by not allowing the materials to move) at the beginning and end of the stitching. Thread tension shall be maintained so that there will be no loose stitching resulting in the puckering of the material sewn. The lock shall be embedded in the materials sewn.

3.4.1.1.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during the stitching, the stitching shall be repaired by restarting the stitching a minimum of 1 inch (1/2 inch for box-x and W-W stitching) from the end of the stitching. 1/

b. Thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the item (in-process or end item) shall be repaired by overstitching. The stitching shall start a minimum of 1 inch from the defective area, (1/2 inch for box-x and W-W stitching), continue over the defective area, and continue a minimum of 1 inch (1/2 inch for box-x and W-W stitching) beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching, without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.4.1.1.2 Repairs of type 308 stitching. Repairs of type 308 stitching shall be in accordance with 3.4.1.1.1a and 3.4.1.1.1b, except that the stitching shall overstitch the defective area a minimum of three stitches onto the existing stitches.

3.4.1.2 Automatic stitching. Automatic stitching machines may be used to perform any of the required stitch patterns provided the requirements for the stitch pattern, stitches per inch, and size and type of thread are met, and at least three or more tying, overlapping, or back stitches are used to secure the ends of stitching.

3.4.2 Thread ends. All thread ends shall be trimmed to a maximum length of 1/4 inch.

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3.4.3 Setting of grommets. Holes shall be prepunched to receive the grommets. Holes prepunched to receive the grommets shall be smaller than the outside diameter of the grommet barrel so that the barrel must be forced through the hole. The grommet shall be securely clinched without cutting the adjacent material.

3.4.4 Finishing of the jacketed wire rope ends. All wire rope ends shall be finished by either the melt method specified in 3.4.4.1 or by the dip method specified in 3.4.4.2 such that the ends are completely enclosed with the melted coating flush with the unmelted nylon jacket.

3.4.4.1 Melt method. Heat the wire rope end nylon jacket with a suitable electrical heating element so as to melt the nylon over the end to form a smooth round surface. Remove any melted residue on the jacket surface with a fine abrasive material such that the surface is not damaged and is finished smooth.

3.4.4.2 Dip method. Coat the wire rope end with a nylon II or nylon 6/6 powder. Heat the coated end for 10 seconds at 405 $\pm$ 15^0F using either an induction coil (2 to 4-inch) with an automatic timer control or an electric melting pot with an automatic temperature control (see 6.4).

3.5 Replacement of defective components. During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects in table I shall be removed from production and replaced with nondefective and properly matched components.

3.6 Repairs. Repairs such as mends, darns, patches, or splices are not permitted on the harness.

3.7 Workmanship. The finished and assembled harnesses shall conform to the quality of the product established by this specification.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)
b. Conformance inspection (see 4.3)

4.2 First article inspection. When a first article inspection is required (see 3.1 and 6.2) it shall be examined in accordance with 4.3. Sampling for first article inspection shall be as specified in the contract or purchase order (see 6.2).

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.3.1,

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4.3.2, 4.3.3 and 4.3.4. Sampling for the inspections in 4.3.3 and 4.3.4 shall be as specified in the contract or purchase order (see 6.2).

4.3.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.2 In-process inspection. Inspection of subassemblies shall be made to ascertain that construction details and dimensional requirements which cannot be examined in the finished product are in accordance with specified requirements. The Government reserves the right to exclude from consideration for acceptance, any material or service for which in-process inspection has indicated nonconformance.

4.3.3 End item visual examination. The end items shall be examined for visual defects. Each noncompliance shall be scored as a defect in accordance with table I.

TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor	
Webbing or tape	Not specified type or class		101	
	Color not as specified			201
	Any hole, cut, or tear		102	
	Edge frayed, beaded, corded, or torn			202
	Ends not seared as specified		103	
Fastener tape	Cut or tear		104	
	Hooks crushed affecting closure		105	
	Any fastener tape reversed, i.e., pile tape attached where hook tape is required or vice versa		106	
	Not specified type and color			203
Cord	Not specified type and color			204
	Ends not seared as specified		107	
Hardware	Fractured, malformed, or bent		108	
	Burrs or sharp edges		109	
	Protective finish omitted or not as specified, any corroded area		110	
	Not specified type, size or style		111	

TABLE I. End item visual defects (contd)

Examine	Defect	Classification

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		Major	Minor
	Adapter attached with friction grip facing wrong direction	112	
Plastic buckle	Not as specified	113	
	Broken, cracked, chipped, or malformed, failing to serve intended purpose; burr, sharp edge or flash	114	
	Color and finish not as specified		205
	Not functioning properly, failing to effect a secure closure or open freely	115	
Jacketed wire rope	Not specified type or size	116	
	Ends not nylon coated or wire material not completely enclosed	117	
	Surface of nylon coated end not smooth or flush with unmelted jacketed surface	118	
Swaged wire rope sleeve	Missing	119	
	Not specified type or size	120	
	Not swaged as specified		206
	One sleeve swage omitted		207
	Two sleeve swages omitted	121	
	Swaged excessively tight, cutting wire rope material	122	
	Swaged loosely, permitting sleeve to rotate freely	123	
Brass grommets	Clinched excessively tight, cutting adjacent material	124	
	Insecurely clinched to a degree that grommet may be detached from material	125	
	Washer installed on incorrect side of material		208
	Barrel split		209

TABLE I. End item visual defects (contd)

Examine	Defect	Classification Major	Minor

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	Loosely clinched, permitting grommet to rotate freely within the hole		210
	Washer omitted	126	
Thread	Not specified type, class, or size	127	
	Not specified color		211
Open seams	1/2 inch or less		212
	More than 1/2 inch	128	
	NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more consecutive skipped or run-off stitches occur.		
Seam and stitch type	Wrong seam or stitch type	129	
Stitches and stitching	Stitching pattern not as specified	130	
	Any row of stitching missing	131	
	One stitch per inch under or over minimum and maximum specified		213
	Two or more stitches per inch under or over minimum and maximum specified	132	
	Thread breaks, bobbin run-outs, or two or more consecutive skipped or run-off stitches not overstitched or restitched as specified	133	
	Ends of stitching not backstitched or secured as specified		214
	Tension loose resulting in loose top or bottom thread, or tight tension causing puckering of material at stitch line	134	
	Needle chews or bunched stitching	135	

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TABLE I. End item visual defects (contd)

Examine	Defect	Classification Major	Minor
Components and assembly	Any component part omitted or not as specified or required operation omitted (unless otherwise classified herein)		136
	Any component misplaced or not assembled as specified		137
	Any darn, splice, or other unauthorized repair		215
Identification marking	Omitted, illegible, incorrect, misplaced, or size of characters not as specified		138
	Color not as specified		216
Cleanness	Dirt, grease, oil or foreign matter clearly noticeable		217
	Thread ends not trimmed to 1/4 inch or less		218
Construction and assembly	Any component missing, or not assembled or joined as specified		139

4.3.4 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified on the drawings. Only those dimensions that can be evaluated without damaging or disassembling the end items shall be examined. Any dimension not within the specified tolerance shall be classified as a defect.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but

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is not mandatory).

6.1 Intended use. The harness is intended for use by individual parachutists for securing individual combat equipment during airborne jump operations. The harness with the secured load is attached to the parachutist's harness D-rings with the attaching strap snaps.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification
b. When first article is required (see 3.1)
c. Sampling plan, lot size, sample size (see 4.3)
d. Packaging requirements (see 5.1)

6.3 Classification deletion. The requirements for type I, "H" Harness have been deleted in this document because the item is no longer procured.

6.4 Source for heat sealing equipment. A source for the heat sealing equipment for forming the ends of the nylon jacketed wire rope (see 3.4.4.2 - induction coil dip method) is Boyd Coatings Research Co., Inc., 51 Parmenter Road, Hudson, MA 01749.

6.5 Subject term (key word) listing.

Airborne jump operations
Combat equipment carrier
Quick release
T-10 parachute harness

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:

Army - GL

Review activity:

Army - AV

Preparing activity:

Army - GL

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